

SECR2043 OPERATING SYSTEMS

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Section : 03

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	Output
	pwd	/home/farra
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a	bash_history .bash_logout .bashrc .local .profile .sudo_as_admin_successful combined.txt file.txt file2.txt filelist.txt hello.txt os_lab practice.txt sample.txt workspace
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir os_lab	No output
4.	Change your current working directory to "os_lab".	
	Command	Output
	cd os_lab	No output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output

	echo "Hello, world!" > hello.txt	No output
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6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	Hello, world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt	No output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt	No output
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt	No output
10.	Display your home directory with additional information, together with subdirectories content.	
	Command	Output
	ls -R /home	/home/farra: total 36 -rw-r--r-- 1 farra farra 158 Apr 24 13:06 combined.txt -rw-r--r-- 1 farra farra 13 Apr 24 12:51 file.txt -rw-r--r-- 1 farra farra 79 Apr 24 13:02 file2.txt -rw-r--r-- 1 farra farra 329 Apr 24 13:03 filelist.txt -rw-r--r-- 1 farra farra 87 Apr 24 15:39 hello.txt drwxr-xr-x 2 farra farra 4096 Apr 25 01:03 os_lab -rw-r--r-- 1 farra farra 759 Apr 24 15:47 practice.txt -rw-r--r-- 1 farra farra 2967 Apr 24 15:46 sample.txt drwxr-xr-x 4 farra farra 4096 Apr 25 01:06 workspace /home/farra/workspace: total 132 -rwxr-xr-x 1 farra farra 16008 Apr 24 16:49 argument -rw-r--r-- 1 farra farra 427 Apr 24 16:49 argument.c -rw-r--r-- 1 farra farra 0 Apr 25 01:04 example.txt -rwxr-xr-x 1 farra farra 15960 Apr 24 16:35 hello -rw-r--r-- 1 farra farra 73 Apr 24 16:34 hello.c -rw-r--r-- 1 farra farra 88 Apr 24 16:43 hello.cpp

		-rwxr-xr-x 1 farra farra 16504 Apr 24 16:44 hello2 -rwxr-xr-x 1 farra farra 16200 Apr 24 16:54 input_keyboard -rw-r--r-- 1 farra farra 482 Apr 24 16:53 input_keyboard.c -rwxr-xr-x 1 farra farra 16144 Apr 24 16:58 kernel_info -rw-r--r-- 1 farra farra 522 Apr 24 16:58 kernel_info.c -rwxr-xr-x 1 farra farra 16008 Apr 24 16:56 multiply -rw-r--r-- 1 farra farra 501 Apr 24 16:56 multiply.c drwxr-xr-x 2 farra farra 4096 Apr 25 06:03 os_lab drwxr-xr-x 2 farra farra 4096 Apr 25 01:04 share
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	No output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World" > hello.txt	No output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	No output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	Hello World This is a test
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	echo "Goodbye World" > bye.txt	No output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	Hello World This is a test Goodbye World
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output

	cat hello.txt bye.txt > greeting.txt	No output
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8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	Hello World This is a test Goodbye World
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	No output
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	Hello farra This is a test Goodbye farra
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	<pre>nano hello.c C program: #include<stdio.h> int main() { printf("Hello World\n"); return 0; }</pre>	No output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	<pre>gcc hello.c -o hello</pre>	No output
3.	Run the executable file "hello" and display its output.	
	Command	Output
	<pre>./hello</pre>	Hello World
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and saves it as "sum.c".	
	Command	Output
	<pre>nano sum.c C program: #include<stdio.h> #include<stdlib.h> int main(int argc, char *argv[]) { if(argc != 3) { printf("Usage: %s num1 num2\n", argv[0]); return 1; } int a = atoi(argv[1]); int b = atoi(argv[2]); printf("Sum = %d\n", a + b); return 0; }</pre>	No output

5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc sum.c -o sum	No output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum 10 20	Sum = 30
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum -5 15	Sum = 10

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	No output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	No output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output

	./echo	Enter a line: Hello World You entered: Hello World
	Total Mark:	